Organic Chemistry Simulator

Documentation and Reagent List

How to Use the Simulator

- **Draw:** Use the Ketcher canvas to draw your reactant molecule.
- **Search:** Click the input box at the top and type to instantly filter available reagents.
- **Reaction Chaining:** You can simulate multi-step synthesis! Type reagents separated by an arrow (`->`). *Example:* `O3 then Zn/H2O -> NaBH4 -> Conc. H2SO4 / Heat`

Alcohols & Phenols

- **HCl / anhy. ZnCl2 (Lucas Reagent):** Alcohols to alkyl chlorides.
- **Conc. H2SO4 / Heat:** Dehydration to alkenes.
- **Na then CH3I:** Williamson ether synthesis.
- **CHCl3 / aq. NaOH:** Reimer-Tiemann (formylates phenol).
- **CO2 / NaOH, then H+:** Kolbe's reaction (carboxylates phenol).
- **Zn dust / Heat:** Reduces phenol to benzene.
- **H+ / Heat (Pinacol Rearrangement):** 1,2-diols to ketones via alkyl shift.
- **NaIO4 or HIO4:** Cleaves 1,2-diols into two carbonyls.
- **PPh3, DEAD, DPPA:** Mitsunobu reaction (alcohol to azide).

Alkyl Halides

- **Aq. KOH / NaOH:** Substitution to alcohols.
- **KCN / AgCN (Alcoholic):** Forms nitriles / isocyanides.
- **KNO2 / AgNO2:** Forms alkyl nitrites / nitroalkanes.
- **NaI / Acetone:** Finkelstein reaction (to Iodides).
- **AgF / Hg2F2 / CoF2:** Swarts reaction (to Fluorides).
- **Alc. KOH / Heat:** E2 Dehydrohalogenation.
- **Mg / Dry Ether:** Forms Grignard reagents.
- **Potassium Phthalimide then Hydrazine:** Gabriel synthesis.
- **(CH3)2CuLi (Gilman Reagent):** Corey-House synthesis.
- **t-BuOK:** Hofmann alkene formation.

Aldehydes & Ketones

- **Zn-Hg / conc. HCl:** Clemmensen reduction (to alkanes).
- **NH2NH2 / KOH, heat:** Wolff-Kishner reduction (to alkanes).
- **Tollens' Reagent:** Oxidizes aldehydes to carboxylates.
- **HCN:** Forms cyanohydrins.
- **I2 / NaOH (Iodoform Test):** Cleaves methyl ketones.
- **Ph3P=CH2 (Wittig Reagent):** Methylenation to terminal alkenes.
- **mCPBA or CF3CO3H:** Baeyer-Villiger oxidation (ketones to esters).
- **Dil. NaOH / Heat (Aldol):** Aldol condensation.
- **Conc. NaOH (Cannizzaro):** Disproportionation of aldehydes.

Alkenes

- **H2 / Pd-C:** Catalytic hydrogenation (syn-addition).
- **Br2 / CCl4:** Anti-addition of bromine.
- **HBr:** Hydrohalogenation.
- **H2O / H+:** Acid-catalyzed hydration.
- **O3 then Zn/H2O:** Reductive ozonolysis.
- **O3 then H2O2:** Oxidative ozonolysis.
- **BH3.THF then H2O2, OH-:** Hydroboration-oxidation.
- **mCPBA (Epoxidation):** Forms epoxides.
- **OsO4 then NaHSO3:** Syn-dihydroxylation.
- **CH2I2 / Zn(Cu):** Simmons-Smith cyclopropanation.

Aromatics & Heterocycles

- **Cl2 / AlCl3 or Br2 / FeBr3:** Halogenation.
- **Conc. HNO3 + H2SO4:** Nitration.
- **Fuming H2SO4:** Sulfonation.
- **CH3Cl / AlCl3:** Friedel-Crafts alkylation.
- **CH3COCl / AlCl3:** Friedel-Crafts acylation.
- **CO / HCl / AlCl3 / CuCl:** Gattermann-Koch formylation.
- **POCl3 / DMF:** Vilsmeier-Haack formylation.
- **Na / Liq. NH3 / EtOH:** Birch Reduction to 1,4-cyclohexadiene.
- **Br2 / Acetic Acid (Heterocycle):** Brominates furan/pyrrole/thiophene at position 2.
- **Fuming HNO3 / H2SO4 / 300°C:** Nitrates pyridine at position 3.
- **mCPBA (Pyridine N-Oxidation):** Forms Pyridine N-oxide.

Carboxylic Acids

- **SOCl2:** Forms acid chlorides.
- **NH3 / Heat:** Forms amides.
- **CH3OH / H+ (Esterification):** Forms methyl esters.
- **Red P / Br2 (HVZ):** Alpha-halogenation.
- **NaOH + CaO / Heat:** Decarboxylation.
- **LiAlH4 then H3O+:** Reduces completely to primary alcohols.
- **Ag2O / Br2 / CCl4 (Hunsdiecker):** Forms alkyl bromides (chain shortening).
- **SOCl2 then CH2N2 then Ag2O/H2O:** Arndt-Eistert homologation.

Alkynes

- **H_2 / Lindlar's Catalyst:** Reduction to cis-alkene.
- **Na / liq. NH_3:** Birch reduction to trans-alkene.
- **H_2 / Ni (excess):** Hydrogenation to alkane.
- **$HgSO_4$ / dil. H_2SO_4:** Kucherov reaction (hydration to ketone).
- **Ammoniacal $AgNO_3$:** Tests for terminal alkynes.
- **Br_2 / CCl_4 (excess):** Halogenation to tetrahalide.
- **Sia_2BH then H_2O_2, OH^-:** Hydroboration to aldehyde.
- **$NaNH_2$ then CH_3I:** Homologation with methyl group.

Advanced Rearrangements

- **NaOH / H_2O (Favorskii):** Alpha-halo ketones to carboxylic acids.
- **KOH / Heat, then H_3O^+ (Benzilic Acid):** 1,2-diketones to alpha-hydroxy acids.
- **Heat / 200°C (Claisen):** Allyl vinyl ethers to gamma,delta-unsaturated ketones.
- **H_2SO_4 / Heat (Beckmann Fragmentation):** Oximes to nitriles and alcohols.

Nitrogen Compounds

- **$CHCl_3$ / KOH (Carbylamine Test):** Primary amines to isocyanides.
- **Br_2 / KOH (Hofmann Bromamide):** Degrades primary amides to primary amines.
- **$NaNO_2$ / HCl, 0-5°C:** Diazotization.
- **Cu_2Cl_2 / HCl (Sandmeyer):** Diazonium to chlorine.
- **HBF_4 / Heat:** Balz-Schiemann reaction (Diazonium to fluorine).
- **H_2O / Warm:** Diazonium to phenol.
- **H_3PO_2 / H_2O:** Mild reduction (deamination).
- **NaN_3 then Heat / H_2O (Curtius):** Acid chloride to primary amine.
- **H_2SO_4 / Heat (Beckmann):** Oxime to amide.

Oxidation & Reduction

- **Acidic $KMnO_4$:** Vigorous oxidation.
- **PCC / CH_2Cl_2:** Mild oxidation (stops at aldehydes).
- **SeO_2 / Heat:** Allylic oxidation.
- **MnO_2 / CH_2Cl_2:** Selectively oxidizes allylic/benzylic alcohols.
- **$NaBH_4$:** Mild reduction (ketones/aldehydes).
- **$LiAlH_4$:** Strong reduction.
- **DIBAL-H / -78°C:** Reduces esters to aldehydes.
- **DIBAL-H then H_3O^+:** Reduces nitriles to aldehydes.
- **$NaBH_4$ / $CeCl_3$:** Luche reduction (1,2-reduction).

Protecting Groups

- **Ethylene Glycol / pTsOH:** Protects carbonyls (acetal).
- **H_3O^+ / Heat:** Deprotects acetals.
- **Boc_2O / Et_3N:** Protects amines (Boc group).
- **TFA / CH_2Cl_2:** Deprotects amines.